

Name: Maths Class:

SYDNEY TECHNICAL HIGH SCHOOL



YEAR 12 HSC COURSE

Mathematics

June 2010

TIME ALLOWED: 70 minutes

Instructions:

- Write your name and class at the top of this page, and on all your answer sheets.
- Hand in your answers attached to the rear of this question sheet.
- All necessary working must be shown. Marks may not be awarded for careless or badly arranged work.
- Marks indicated are a guide only and may be varied at the time of marking

(FOR MARKERS USE ONLY)

1	2	3	4	5	TOTAL
/12	/10	/11	/10	/11	/54

QUESTION 1: (12 Marks)

	Marks
(a) Find $\int \frac{dx}{\sqrt{x}}$	2
(b) Give the exact value of $\int_0^{\frac{\pi}{8}} \sec^2 2x dx$	3
(c) Give the exact value of $\cos \frac{7\pi}{4}$	1
(d) Solve the equation $\tan x = -\sqrt{3}$ for all values of x in the Domain $0 \leq x \leq \pi$	2
(e) The minute hand of a clock is 3 cm in length. What area is swept out by the minute hand over a 40 minute period?	2
(f) Find $\frac{d}{dx} \sin(3x^2 + 5)$	2

QUESTION 2: (10 Marks)

	Marks
You are given the curve $y = x^3 - 9x$	
(i) Where does this curve cut the x -axis?	1
(ii) Find all stationary points on the curve, and determine their nature.	4
(iii) Find the point of inflexion.	2
(iv) Sketch the curve showing all features you have just found.	3

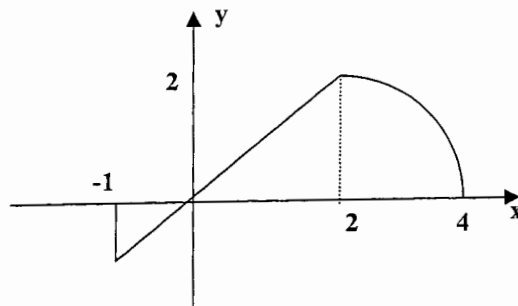
QUESTION 3: (11 Marks)

Marks

- (a) The graph of $y = f(x)$ for some function $f(x)$, is drawn below:

3

Use this graph to find the value of $\int_{-1}^4 f(x) dx$



- (b) The value of a certain function $g(x)$ for values of x from 1 to 5 is given in the table below:

3

x	1	2	3	4	5
$g(x)$	1	0.8	1.5	2	1.6

Using Simpson's Rule with 5 function values, find an approximation for the area enclosed by the curve $y = g(x)$, the x -axis and the lines $x=1$ and $x=5$

- (c) You are given that $\int_1^a \frac{dx}{x^2} = \frac{1}{2}$

2

Find the value of a .

- (d) Find the area between the curve $y = 4 - x^2$ and the x -axis.

3

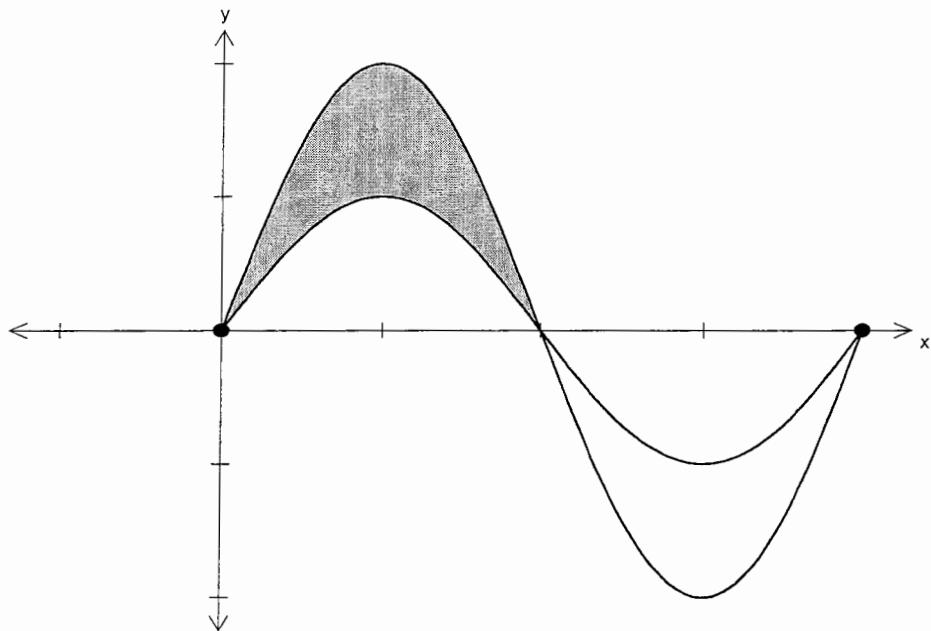
QUESTION 4: (10 Marks):

Marks

- (a) An open water tank has a square base of side x m and vertical sides. It is to be built out of 75m^2 sheet metal (the base is also metal)

- (i) If V is the volume of the water tank, show that $V = \frac{x}{4}(75 - x^2)$ 2
- (ii) Find the dimensions of the tank if it is to hold as much water as possible. 4

- (b) (i) Rough sketches of the curves $y = 2\sin 2x$ and $y = \sin 2x$ are drawn below. *Neatly copy this diagram onto your answer sheet and complete the labelling of both axes* 2



- (ii) Find the value of the shaded area. 2

QUESTION 5: (11 Marks)**Marks**

- (a)
- (i) Draw the curve $y = \tan x$ for the Domain $-\frac{\pi}{2} \leq x \leq \frac{\pi}{2}$ 2
- (ii) Find the equation of the tangent to the curve $y = \tan x$ at the point where $x = \frac{\pi}{4}$ 3
- (iii) By considering your diagram, how many solutions are there of the equation 1
- $$\tan x = 2x + 1 - \frac{\pi}{2} \quad \text{in the domain } -\frac{\pi}{2} \leq x \leq \frac{\pi}{2}$$

- (b)
- (i) Show that $\int_0^3 (x+1)^{\frac{3}{2}} dx = \frac{62}{5}$ 2
- (ii) Explain, algebraically, why $(x+1)^{\frac{3}{2}} = x\sqrt{x+1} + \sqrt{x+1}$ 1
- (iii) Using your answers to parts (i) and (ii) above, find the value of 2

$$\int_0^3 x\sqrt{x+1} dx$$

END OF EXAMINATION PAPER

2 UNIT SOLUTIONS
(and MARKING)

QUESTION 1: (12 MARKS)

(a) $\int \frac{dx}{\sqrt{x}} = 2x^{\frac{1}{2}} + k$

2 MARKS This is the only place in the paper where mark is deducted for no "k"

(b) $\int_0^{\frac{\pi}{2}} \sec^2 2x dx = \left[\frac{1}{2} \tan 2x \right]_0^{\frac{\pi}{2}}$

← ① MARK

$= \frac{1}{2} (\tan \frac{\pi}{4} - \tan 0)$

← ① MARK

$= \frac{1}{2}$

← ① MARK

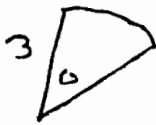
(c) $\cos \frac{7\pi}{4} = \cos \frac{\pi}{4}$
 $= \frac{1}{\sqrt{2}}$

1 MARK

(d) $x = \frac{\pi}{3}, 2\frac{\pi}{3}, 5\frac{\pi}{3}$

1 MARK each part.

(e)



$40 \text{ units} = \frac{2}{3} \text{ circ.}$

$\therefore \theta = \frac{4\pi}{3}$

1 MARK for $\frac{4\pi}{3}$

$\therefore A = \frac{1}{2} (3)^2 \cdot \frac{4\pi}{3}$

$= 6\pi \text{ cm}^2$

1 MARK for answer

(f) $\frac{d}{dx} \sin(3x^2 + 5) = 6x \cos(3x^2 + 5)$

1 for cos

1 for $6x$.

QUESTION 2. (10 MARKS)

$$y = x^3 - 9x$$

(i) cuts at $(0,0)$ and $(3,0)$ and $(-3,0)$ 1 ONLY

(ii) $\frac{dy}{dx} = 3x^2 - 9$

$$\frac{d^2y}{dx^2} = 6x$$

At T.P.'s $\frac{dy}{dx} = 0$

$$\Rightarrow x = \sqrt{3} \text{ or } x = -\sqrt{3}$$

$$y = +6\sqrt{3}$$

$$y'' \geq 0$$

$$\Rightarrow \text{min T.P. at } (\sqrt{3}, -6\sqrt{3})$$

$$y = -6\sqrt{3}$$

$$y'' \leq 0$$

$$\Rightarrow \text{max T.P. at } (-\sqrt{3}, 6\sqrt{3})$$

1 for each point
1 for each max/min
(ie. 4)

(iii) at I.P.'s $\frac{d^2y}{dx^2} = 0$

$$\Rightarrow \begin{cases} x = 0 \\ y = 0 \end{cases} \quad \begin{array}{c} -ve \quad 0 \quad +ve \\ - \quad 0 \quad + \end{array}$$

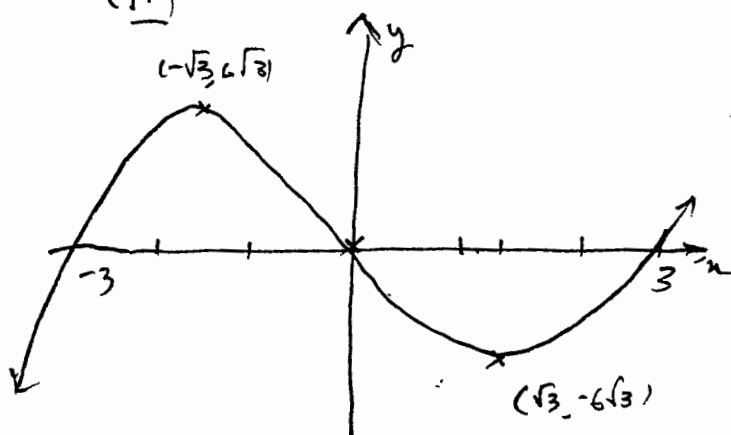
2nd derivative changes sign.

$\therefore$ I.P. at $(0,0)$

1 for $(0,0)$

1 for derivative
changing sign

(iv)



① for x-axis
intercepts

① for T.P.'s

① for I.P. at
 $(0,0)$

QUESTION 3:

$$\begin{aligned} \underline{(a)} \quad A_1 &= \frac{1}{2}(1 \times 1) = \frac{1}{2} \\ A_2 &= \frac{1}{2} \times 2 \times 2 = 1 \\ A_3 &= \frac{1}{4} \times \pi \times (2)^2 \\ &= \pi \end{aligned}$$

$\therefore$ ~~Area~~

$$\begin{aligned} \therefore \text{Integral} &= -\frac{1}{2} + 1 + \pi \\ &= \pi + \frac{1}{2} \end{aligned}$$

$$\begin{aligned} \underline{(b)} \quad A_1 &= \frac{1}{3} [1 + 1.5 + 4 \times 0.8] = 1.9 \\ A_2 &= \frac{1}{3} [1.5 + 1.6 + 4 \times 2] = 3.7 \\ \therefore A &\approx 5.6 \text{ m}^2 \end{aligned}$$

$$\begin{aligned} \underline{(c)} \quad \int_1^a \frac{dx}{x^2} &= \left[-x^{-1} \right]_1^a \\ &= -\frac{1}{a} + 1 = \frac{1}{2} \\ \therefore \frac{1}{a} &= \frac{1}{2} \\ \therefore a &= 2 \end{aligned}$$

(d) Cuts x-axis at (2,0) and (-2,0)

$$\begin{aligned} \therefore A &= \int_{-2}^2 (4 - x^2) dx \\ &= \left[4x - \frac{1}{3}x^3 \right]_{-2}^2 \\ &= 8 - \frac{8}{3} - \left(-2 + \frac{8}{3} \right) \\ &= \frac{32}{3} \text{ m}^2 \end{aligned}$$

3 marks

(2 marks) for the areas
plus (1) for adding two and subtracting one area
(ie only (2) for $\pi + \frac{3}{2}$)

1 for each part
or
There are other S.R. formulae

← (1) for correct integral

← (1) for $a=2$

(1) for limits of $x=2, x=-2$

(1) for correct integration

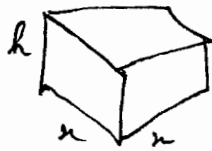
(1) for answer

CAN ALSO BE DONE

by $2 \int_0^2 (4 - x^2) dx$.

QUESTION 4:

(a)



(i) $SA = x^2 + 4xh = 75$

$$VOL = x^2 h$$

$$= x^2 \left(\frac{75 - x^2}{4x} \right)$$

$$= \frac{x}{4} (75 - x^2)$$

(ii) $\frac{dV}{dx} = \frac{75}{4} - \frac{3x^2}{4}$

$$\frac{d^2V}{dx^2} = -\frac{6x}{4}$$
$$= -\frac{3x}{2}$$

At max vol, $\frac{dV}{dx} = 0$

$$\Rightarrow 3x^2 = 75$$

$$\begin{cases} x = 5 \text{ or } -5 \end{cases}$$

$$h = \frac{5}{2}$$

$$V = \frac{125}{2}$$

$$V'' < 0 \Rightarrow \text{max.}$$

$\therefore$ Tank is $5 \times 5 \times \frac{5}{2}$ m.

① for getting

$$h = \frac{75 - x^2}{4x}$$

① for substituting into volume formula

} ① for both $\frac{dV}{dx}$ and $\frac{d^2V}{dx^2}$

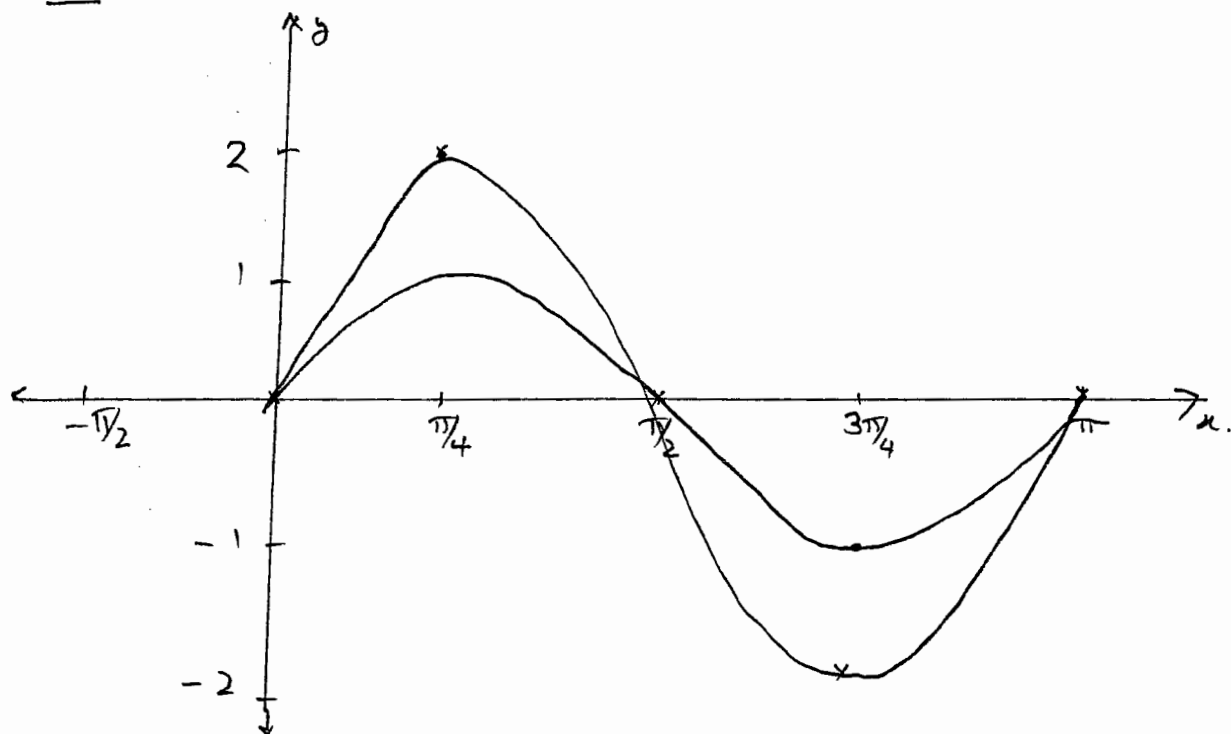
← ① for recognising this

~~for~~

① for testing MAX

① for dimensions

4(b)



2 MARKS

① for vertical axis labelling

① for horizontal axis labelling

[DON'T BE TOO TOUGH - as long as they give the idea they know what the dimensions are!]

(ii)

$$A = \int_0^{\pi/2} 2\sin 2x - \sin x \, dx \quad \leftarrow \text{① for this line}$$

$$= \left[-\frac{1}{2} \cos 2x \right]_0^{\pi/2}$$

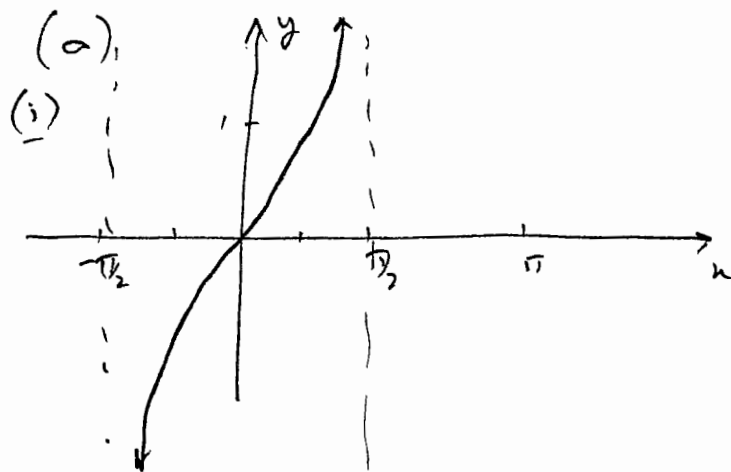
$$= -\frac{1}{2} \cos \pi + \frac{1}{2} \cos 0$$

$$= 1 - \frac{1}{2}$$

$$= \frac{1}{2} \text{ u}^2$$

① for answer

QUESTION 5:



① for asymptotes

① for shape

(ii) $\frac{dy}{dx} = \sec^2 x$

← ①

At $x = \pi/4$ $m_T = \sec^2 \pi/4$
 $= 2$

← ①

∴ Equation is

$$y - 1 = 2(x - \pi/4)$$

$$\therefore y = 2x - \pi/2 + 1$$

← ①

(iii)

TWO

← ① RIGHT OR WRONG.

(b) (i) $\int_0^3 (x+1)^{3/2} dx = \left[\frac{2}{5} (x+1)^{5/2} \right]_0^3$
 $= \frac{2}{5} (4)^{5/2} - \frac{2}{5}$
 $= \frac{62}{5}$

← ① for integral

} ① for powers

(ii) $(x+1)^{3/2} = (x+1)\sqrt{x+1}$
 $= x\sqrt{x+1} + \sqrt{x+1}$

} anything acceptable = ①

(iii) $\int_0^3 x\sqrt{x+1} dx = \int_0^3 (x+1)^{3/2} dx - \int_0^3 \sqrt{x+1} dx$ ← ① for recognising this
 $= \frac{62}{5} - \left[\frac{2}{3} (x+1)^{3/2} \right]_0^3$
 $= \frac{62}{5} - \frac{16}{3} + \frac{2}{3} = \frac{26}{15}$ ← ①